Following the Bottleneck: A Lightweight Solution to the IEEE BigData 2026 Cup TrafficFlowBench

Kuan-Ting Kuo
Team Anton
kuo.kuant@gmail.com
Code: https://github.com/kuokuant-oss/Lightweight-Solution-to-the-IEEE-BigData-2026-Cup-TrafficFlowBench-Competition

***Abstract*—TrafficFlowBench asks for freeway speed and flow where detectors are missing, for 30-min forecasts of which links will be queued, and for path flows, on ten corridors whose evaluation months are generated independently of the training months. Our solution follows the freeway bottleneck, whose mechanism holds in every month and whose first breakdown moves from month to month. A bottleneck discharges at capacity and breaks down with the phase of 30-min demand steps, so a new queue is forecast from each bottleneck's breakdown risk, a conditional-logit choice of the bottleneck that breaks first, and that bottleneck's queue extent. Adjacent links enter and leave a queue in fixed co-switching groups, which serve as one spatial unit across imputation, gap reconstruction and most queue decisions. Each forecast origin is followed by 90 min in which the whole corridor is unobserved; a gap model reads the interval's boundaries, and the on-ramp counts that stay visible give each group a queue probability for a two-state speed. Gradient-boosted trees and small regressions carry the whole solution, which trains and runs on a laptop CPU with about 0.5 GB of models and tables. The final submission scores 0.901 public and 0.895 private and ranks 12th on the private leaderboard.**

***Index Terms*—traffic state estimation, queue forecasting, freeway bottleneck breakdown, co-switching link groups, physics-informed machine learning, conditional logit, distribution shift, lightweight gradient boosting**

1. Introduction

When loop detectors fail, a traffic management centre needs the speed and flow on every link, a state that conserves vehicles, an early warning of where queues will form or persist, and path flows that agree with the counts. TrafficFlowBench [1], run as a Kaggle competition [2], scores these four needs as state reconstruction (Task 1), queue forecasting (Task 2), physical consistency (Task 3) and path-flow estimation (Task 4). The data are synthetic 5-min detector records of mainline speed, flow and occupancy, with on- and off-ramp counts, on ten directional freeway corridors in California: nine fully observed training months, followed by two evaluation months released with masks.

The evaluation months are generated independently of the training months. In March and April queues often start at other bottlenecks, and on corridors with several recurrent bottlenecks any of them can break first, so where queues formed is knowledge specific to a month. Every queue-forecast origin is followed by a 90-min interval in which the whole corridor's mainline is withheld, and the Task 1 cells inside it are scored while all their same-time neighbours are withheld.

Data-driven and physics-informed state estimation [3] and low-rank imputation [4] are well developed. Closest to our work is the bottleneck-centred view of congestion prediction: Yildirimoglu and Geroliminis [5] locate the active bottlenecks and predict the extent of their congestion from stochastic congestion maps, and later work models the occurrence of congestion patterns with discrete choice [6]. Our contributions extend this view:

1) **A bottleneck mechanism that holds across months, turned into a choice model.** A logistic model per bottleneck estimates breakdown risk from the flow-to-capacity ratio and the phase of the 30-min demand step, two quantities that keep their meaning from month to month; a conditional logit compares the candidate bottlenecks within the forecast window and picks the one that breaks first; the chosen bottleneck's queue-extent template recovers the queued links.
2) **Co-switching link groups as one spatial unit.** Links that enter and leave a queue together form groups that serve imputation, the reconstruction of blanked intervals and the decision of link-level queue forecasts on most corridors; the same idea, applied to each month's own observations, shares the queue probability inside blanked intervals.
3) **Boundary-based reconstruction of corridor-wide blanked intervals.** A gap model built on the interval's boundaries gave the largest gain of the development; the on-ramp counts visible inside the interval refine its speed with one queue probability per group, which weights a two-state speed.
4) **A lightweight, reproducible pipeline.** Every model is trained and served on a laptop CPU, its models and tables fit in about 0.5 GB, and one script reproduces the submitted file exactly.

2. Data and Findings

2.1 Split and scoring

The training period, June 2030 to February 2031, is fully observed; March 2031 feeds the public and April 2031 the private leaderboard. We develop on June to January, keep February as an in-period holdout, whose queues form at the same places as in the other training months, and test transfer on the released, partly masked March observations (Section 3.1); several final

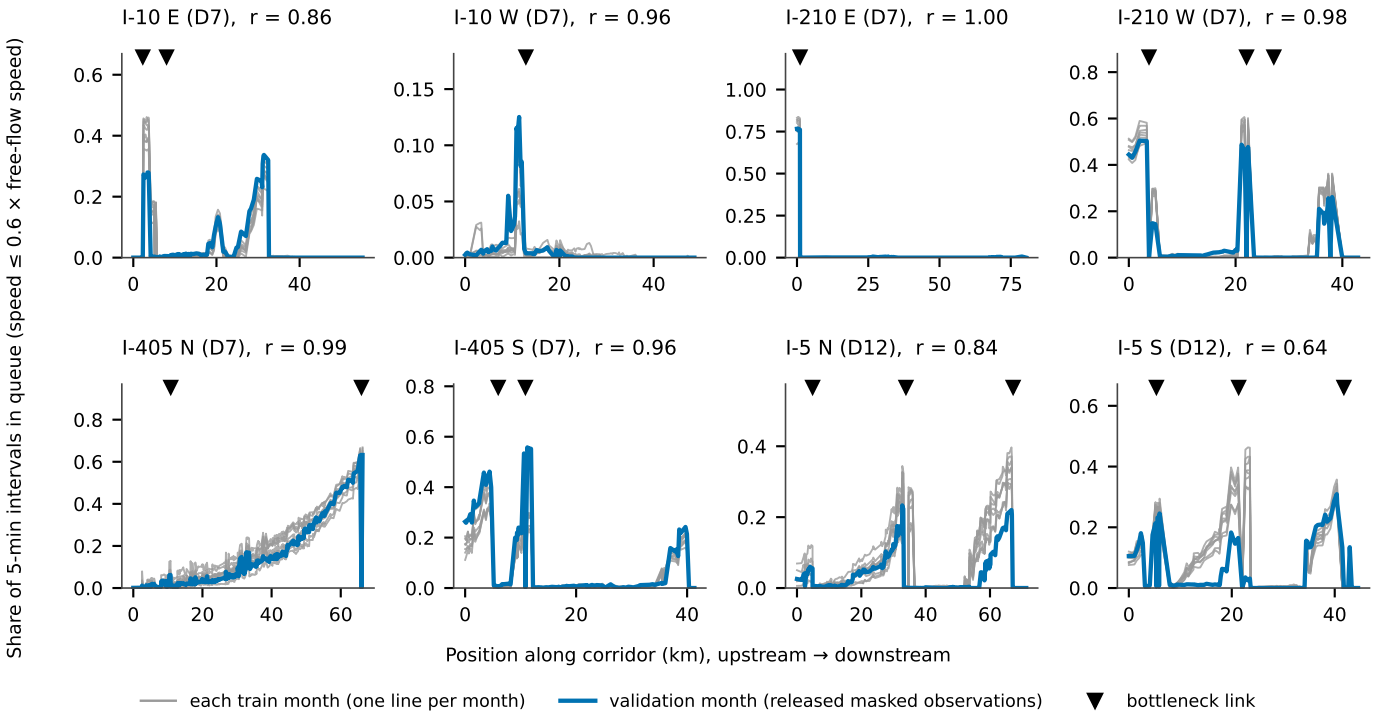


Fig. 1. Share of weekday 5-min intervals in queue along each Task 2 corridor, 05:00–21:00. Grey: training months; blue: the validation month, March (released masked observations). r: correlation of the March link profile with the mean training profile; training months correlate with one another at 0.95 or higher. ▼: bottleneck links (two within 1.5 km drawn once). D7 and D12: Caltrans districts.

models are refitted through February (Appendix C). A *cell* is one link in one 5-min interval, and a link is *queued* at speeds up to the benchmark's cut-off speed of $0.6\,v_{\mathrm{f}}$; the free-flow speed v_{f} and the capacity c of each link are constants published by the organisers. The total score S_{total} weights the state, queue, physics and path-flow scores (S_{state}, S_{queue}, S_{physics}, S_{ODME}, the last for origin–destination matrix estimation) by 0.35, 0.30, 0.15 and 0.20. The queue score averages a space–time intersection over union (IoU) per forecast window, so one window moving from IoU 0 to 1 is worth 0.00375 of S_{total}. The organisers select the scored windows by a published rule: every queue-onset window is queue-free until $T{+}30$, and every ongoing-queue window sits where persisting the current queue would score an IoU of at most 0.9, that is, at a turning point of the queue.

2.2 Three findings that shaped the models

Finding 1: corridor-wide blanked intervals remove every cell's spatial neighbours. In the evaluation months, the corridor's mainline is withheld for 90 min after each queue-forecast origin, the 30-min horizon plus one hour. These cells are 1.6–1.7% of the Task 1 targets, and all their same-time neighbours are withheld, a block-missing pattern; the training months hold scattered missing cells. At one bottleneck, 77% of the vehicles that crossed it during an interval were missing from our first reconstruction.

Finding 2: the first bottleneck to break changes between months; the bottleneck mechanism is stable. The link-level

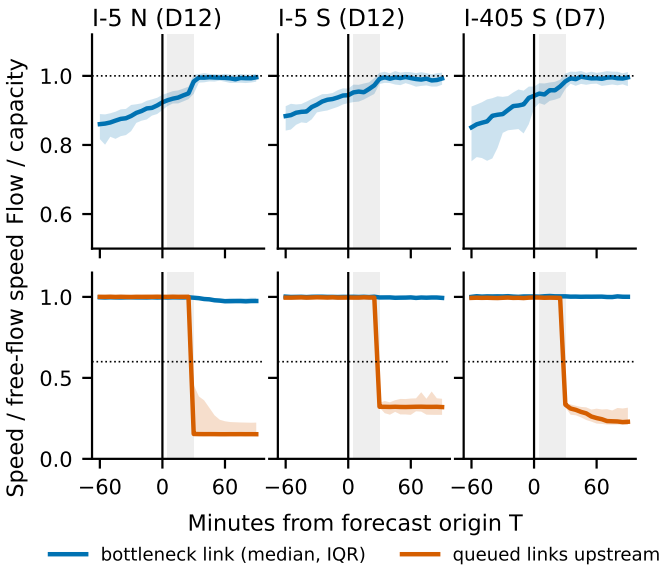


Fig. 2. Bottleneck link around training onsets on three multi-bottleneck corridors, $T{-}60$ to $T{+}90$ (onset at $T{+}30$, grey: horizon). Top: flow-to-capacity ratio; bottom: speed relative to free-flow speed, with the queued links upstream. Median and interquartile band; dotted lines: capacity (top) and the queue threshold $0.6\,v_{\mathrm{f}}$ (bottom).

queue profiles of the training months correlate with one another at 0.95 or higher, and those of March with them at 0.64–1.00 (Fig. 1). On one corridor, the bottleneck that started 62% of the onsets in the training months started none

in March. The mechanism holds in every month (Fig. 2). While a queue is present, the bottleneck link discharges at capacity at free-flow speed, with the queue upstream. Demand changes in 30-min steps at the same minute in March as in training, and 61% of breakdowns fall in the step interval or the next one, against 33% if uniform. Breakdown is a probabilistic event whose chance rises with flow relative to capacity [7]. Once a bottleneck breaks, its template links queue in 97–100% of its onsets.

Finding 3: queues switch in fixed groups of links. For each pair of adjacent links, we measure the share of the times one link enters the queue in the training months in which the other enters in the same 5-min step. Of the 614 pairs with at least 20 such entries, 44% exceed 0.8 and 42% fall below 0.2, so a cut anywhere between 0.2 and 0.8 moves at most 14% of the pair boundaries. Cutting at 0.5 gives 47–170 contiguous co-switching groups per corridor, and each group enters and leaves the queue as one unit, with a purity of 0.997 in March.

3. Method

3.1 Validation protocols

Our validation protocols reproduce the evaluation conditions, so that a local score reflects the blanked intervals and the month-to-month distribution shift. For Tasks 1 and 3, the rule-replicated gap protocol cuts 90-min corridor-wide intervals in the training months at origins chosen by the window-selection

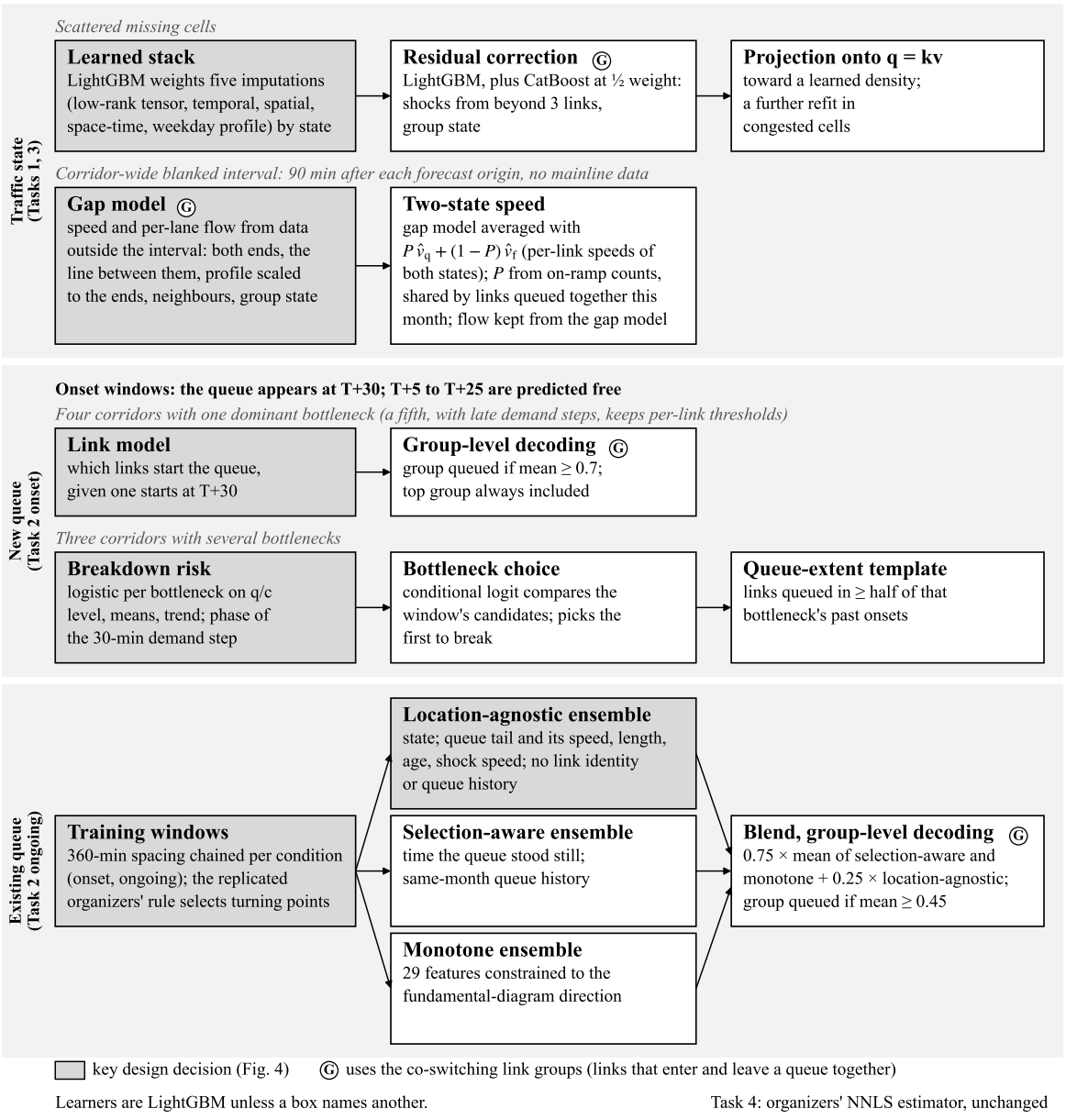


Fig. 3. Method overview. Each lane is one scored problem; italic labels give the cells or corridors each row serves. Shaded boxes are the six key design decisions of Fig. 4. q/c: flow-to-capacity ratio; $\hat{v}_q$, $\hat{v}_f$: predicted speeds of the queued and the free state.

rule and scores blanked and scattered missing cells together. For Task 2, out-of-period evaluation uses replicated forecast windows: windows cut from the released March observations with a replicated selection rule, which reproduces 75 of the 80 official training windows, scored on the observed speeds with the same code that serves the submission. Labels are observed cells outside blanked intervals, and origins whose history touched a blanked interval were skipped. Replicated histories are masked more heavily than official ones (about 41% against 16% missing), so some validation runs filled their history cells with our offline Task 1 reconstruction; this fill belongs to validation. Task 2 itself is served online: for a window with origin T, it reads released data stamped at or before T and models fitted on the training months, and no Task 1 output (Appendix B).

3.2 Baseline

Our first complete submission assembled methods from the literature and scored 0.789 public and 0.807 private. Task 1 blended five classical reconstruction members, low-rank tensor completion [4], temporal, spatial and space–time interpolation, and a weekday×time profile, by non-negative least squares (NNLS), as in stacked regressions [8]. Task 2 used LightGBM [9] link classifiers that start from each link's historical queue frequency, following a Traffic4cast solution [10], with separate models for onset and ongoing windows, training windows cut from the training months by the organisers' rule, and equal weight per window as in the IoU score. Task 3 was scored on the Task 1 output as it stood, since networks trained with conservation terms in the loss misplace the shocks at queue boundaries [11]. Task 4 kept the organisers' prior-regularised estimator. LightGBM remains the main learner, and small models support it where a traffic judgement has a simple form (Table I). Six decisions, shaded in Fig. 3, changed this baseline.

TABLE I
LEARNERS: WHAT EACH IS USED FOR AND WHY IT WAS KEPT.

Learner	Used for	Why this learner
Low-rank tensor completion [4]	One of five reconstruction members	Reads day-to-day similarity and complements interpolation (residual correlation 0.22).
LightGBM	Stack, residuals, density, gap model, queue classifiers	Tabular traffic state; the strongest queue learner against a sequence network, a temporal-convolution encoder and CatBoost.
CatBoost [12]	Second residual learner	Errs on different cells; half weight keeps the conservation gain.
Logistic regression	Breakdown risk per bottleneck	Few features with a known direction; one small model per bottleneck.
Conditional logit [13]	First bottleneck to break	Breakdown order is decided by comparing bottlenecks at the same moment.
NNLS	Path flows (organisers' estimator)	The prior keeps total demand where the counts support it.

3.3 Traffic state (Tasks 1 and 3)

Learned stack for scattered cells. The baseline blend applies one weight set to every cell, so queued and free-flowing cells share weights; on one corridor, the 10.6% of cells below 60 km/h produced 55.5% of the conservation error. A LightGBM stack learns which member to trust in each traffic state from the members and the local context. A residual correction adds information from beyond three links: shocks propagating from far along the corridor, read along the propagation direction as in adaptive smoothing [14], and the state of the rest of the link's co-switching group. Physics enters after learning: a weighted least-squares step moves each reconstructed cell's speed and flow onto $q = k\,v$, pulling the density toward a separately learned one, with a further refit below $0.85\,v_{\mathrm{f}}$. This step outscored a conservation term in the loss and temporal smoothing.

Gap model for the blanked intervals. The intervals of Finding 1 are reconstructed by a pair of LightGBM regressors, for speed and per-lane flow, whose inputs all come from outside the interval: the observations at both ends, the interpolation between them, a weekday profile rescaled by the boundary values, neighbouring links, and the state of the cell's co-switching group before and after the interval. Training intervals are cut from the training months by the same rule. On February intervals, the root-mean-square error (RMSE) of speed fell from 10.1 km/h for linear interpolation to 5.4 km/h.

The queue boundary inside the interval moves from month to month, and the cells that switch state carry 60% of the speed error. The interval speed therefore averages the gap model with a two-state expectation, $v = P\,\hat{v}_{\mathrm{q}} + (1-P)\,\hat{v}_{\mathrm{f}}$, where $\hat{v}_{\mathrm{q}}$ and $\hat{v}_{\mathrm{f}}$ are each link's speeds predicted for the queued and the free state by two LightGBM heads; this expectation is the minimum-squared-error estimate when the state is uncertain. The queue probability P comes from a LightGBM classifier on the boundary values and the on-ramp counts that remain visible inside the interval, because they fall sharply when the mainline queue spills back to the ramp. P is averaged over links whose queues coincided on other days of the same month, which lowered the interval speed error more than averaging over the training-month groups. The heads and the classifier are fitted on intervals cut into each month's released observations, and flow stays with the gap model.

3.4 New queues (Task 2 onset)

Link model with onset-conditional incidence. For links that rarely queued, the baseline's historical frequency, clipped at a small floor, fixed a large negative starting log-odds that dominated every tree correction, so onsets at new times and places received near-zero probability. The model now estimates which links start the queue, conditional on a queue starting at $T{+}30$: the frequency becomes an ordinary input, joined by the incidence of each link and time of day over training onsets, pooled toward link and global rates. On the four corridors with one dominant bottleneck, the probabilities are averaged within each co-switching group (Finding 3),

and a group is queued when the mean reaches 0.7, with the most probable group always included; 0.7 gained more on the replicated March windows than 0.5 or 0.6. A fifth corridor, with several bottlenecks and large demand steps after the origin, keeps the link model with per-link thresholds. Since onset windows are queue-free until $T{+}30$ (Section 2.1), steps $T{+}5$ to $T{+}25$ are predicted free, and every window receives at least its most probable link.

Risk, choice and extent on multi-bottleneck corridors. On the other three corridors, two to four bottlenecks start onsets, and most onset loss came from the choice among them: on the replicated March windows, 20 of 155 windows produced 80% of it. Training-month frequency follows the old months (Finding 2), and bottlenecks break at flow-to-capacity ratios from about 0.90 to 0.98, so each bottleneck gets its own model. *Risk:* one standardised logistic regression per bottleneck predicts breakdown at $T{+}30$ from the bottleneck link's flow-to-capacity ratio q/c at T, its 15-, 30- and 60-min means, its recent maximum and increase, and the phase of $T{+}30$ in the demand step. *Choice:* the order of breakdown is decided by comparing bottlenecks at the same moment, so a conditional logit treats each window w as a choice among its candidates $\mathcal{B}_w$, the bottlenecks with at least 10 training onsets,

$$P(j \mid w) = \frac{\exp(\beta^\top x_{jw})}{\sum_{k \in \mathcal{B}_w} \exp(\beta^\top x_{kw})}, \tag{1}$$

where x_{jw} holds bottleneck j's q/c level, trend and demand-step phase, their differences and ranks across the candidates, and its logistic risk, and the coefficients β are fitted on training onsets; we average 24 fitted logits. When every candidate's risk is below 0.01, the window falls back to the link model. *Extent:* the chosen bottleneck's template marks the links queued in at least half of its training onsets; template links queue in 97–100% of the bottleneck's onsets (Finding 2), so this cut matters little. Top-1 bottleneck accuracy on the replicated March windows rose from 0.46–0.69 with training-month frequency to 0.78 with the risk models, and the conditional logit raised it further.

3.5 Existing queues (Task 2 ongoing)

Training windows that look like the scored ones. The selection rule keeps forecast origins at least 360 min apart. Our first sampler applied this spacing as one chain over both conditions, so the hours after an onset window contributed no ongoing windows: on one corridor, 2,570 eligible ongoing origins yielded 2 windows. One chain per condition restored them, and windows cut by the replicated selection rule then matched the scored ones at the turning points of the queue.

A location-agnostic ensemble. A model with link identity and historical queue frequency extends queues as they extended in the training months: on the replicated March windows it predicted extensions on 4.7% of cells where none occurred, against 1.4% in training. The location-agnostic ensemble, six LightGBM classifiers, reads the current state in place of these two inputs: the state of the link and its neighbours at 1, 2 and 4 links, ramp flows, time of day, and the geometry of the visible queue, namely tail position and speed, length, age, and the shock speed between arriving flow and queue [15]. Each recurrent queue's tail moves at a stable speed, so its recent motion locates the tail 30 min later from the state alone.

Two ensembles add complementary information. A selection-aware ensemble reads how long the queue has been stationary, from the number of consecutive earlier origins that the selection rule skipped because persistence was still accurate, and the link's queue share on earlier days of the same month. A monotone ensemble constrains 29 state features to the direction of the congested branch of the fundamental diagram. The link probability weights the mean of these two ensembles by 0.75 and the location-agnostic ensemble by 0.25; it is averaged within each co-switching group and decided at 0.45, chosen among 0.40–0.60 on odd replicated days and confirmed on even days. Group-level decoding outscored link thresholds and expected-IoU decoders that treat cells as independent on the replicated March windows.

3.6 Physical consistency and path flows (Tasks 3 and 4)

For Task 3, the auxiliary density model behind the projection of Section 3.3 is trained with an L1 loss weighted by link length, which directly minimises the vehicle-count error of the missing cells; reconstructed cells already sit close to the fundamental diagram, so this conservation error is where the physics score gains.

For Task 4, we keep the organisers' estimator $\min_{f \geq 0} \|Af - c\|^2 + 0.05\,\|f - b\|^2$ with the released prior b, since 35–100 independent counts constrain 630–5,050 paths per corridor. A paired submission placed S_{ODME} at about 0.994, and alternative penalties and priors shifted the total demand in directions the counts leave open.

3.7 Leaderboard effect of the decisions

From the baseline to the final submission (Table II), S_{total} rose by $+0.112$ public and $+0.088$ private, reaching 12th on the private leaderboard. The six decisions of Fig. 4 account for $+0.073$ public and $+0.072$ private, and all other steps together for $+0.039$ and $+0.016$, led on public by the onset choice and template (Appendix A).

4. Efficiency and Reproducibility

Recent deep imputers, such as diffusion [16] and graph models [17], are trained on GPUs; diffusion imputers add tens of denoising steps per sample, and network-scale graph forecasters report more than 100 GPU hours of training [18]. Our solution takes the tabular route, in line with evidence that tree ensembles stay competitive on tabular data at lower computational cost [19], and keeps each traffic judgement in a small model; survival models with Weibull capacity [7] are a richer form of the breakdown component.

The recorded fitting runs of the frozen models and tables add up to about 3 h on a laptop CPU (Apple M4, 10 cores, 16 GB RAM). One script reruns the pipeline from the released data and the frozen, checksummed models and tables in 70 min

TABLE II

FINAL SUBMISSION BY TASK. TASK 2 SCORES COME FROM PAIRED SUBMISSIONS THAT BLANK ALL QUEUE CELLS, OR THE ONSET CELLS; TASKS 1, 3 AND 4 ARE REPORTED TOGETHER.

	Public	Private
S_{total}	**0.9012**	**0.8949**
S_{queue} (Task 2)	0.866	0.848
Onset IoU	0.873	0.789
Ongoing IoU	0.858	0.908
$0.35\,S_{\text{state}} + 0.15\,S_{\text{physics}} + 0.20\,S_{\text{ODME}}$	0.641	0.640

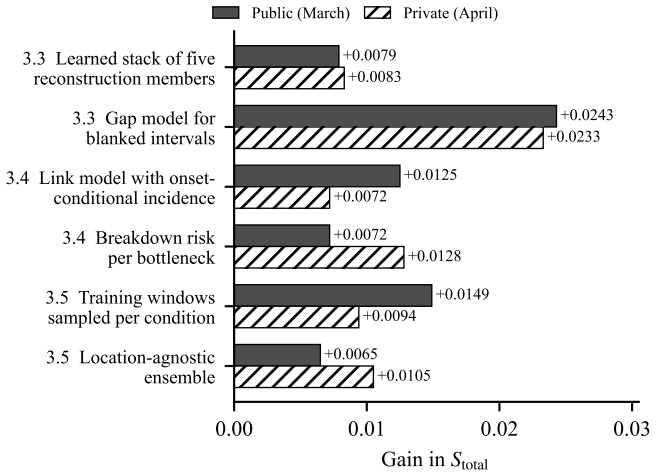


Fig. 4. Leaderboard gain of the six design decisions, in the order of Section 3, each read from the submission that introduced it. Three of those submissions also changed another part: Task 1 flow on two corridors (onset-conditional incidence), group-level decoding for both queue conditions (breakdown risk), and density projection on five corridors and four onset cells (training windows).

(Table III); Task 2 fits no model at serving and reruns on a new month in the released layout in about 5 min (Appendix B). Task 1 takes most of the 70 min, since it completes each month's tensor and refits the gap-speed correction, the two-state heads and the queue classifier on each month's released observations. The rerun reproduced the submitted file byte for byte (SHA256), and the package records each model's training data, features, parameters and seeds (Appendix C). Lightweight here means that training and inference both run on one laptop CPU, with about 0.5 GB of models and tables and a peak memory of 5.5 GB.

5. DISCUSSION AND LIMITATIONS

5.1 Decisions that read the current state transfer across months

The evaluation months move queues to new bottlenecks, and the decisions that read the current state, the breakdown-risk models and the location-agnostic ensemble, gained more on the private month than on the public one. Validation follows the same pattern: on the in-period holdout, the ongoing-queue model with link identity and historical frequency ranked

TABLE III

WALL TIME OF A FULL RERUN ON A LAPTOP CPU (APPLE M4, 10 CORES, 16 GB RAM).

Stage	Wall time
Traffic state (Tasks 1 and 3)	54 min
New queues (Task 2 onset)	1 min
Existing queues (Task 2 ongoing)	5 min
Path flows (Task 4)	10 min
Staging and assembly	<1 min
Total (peak memory 5.5 GB)	70 min

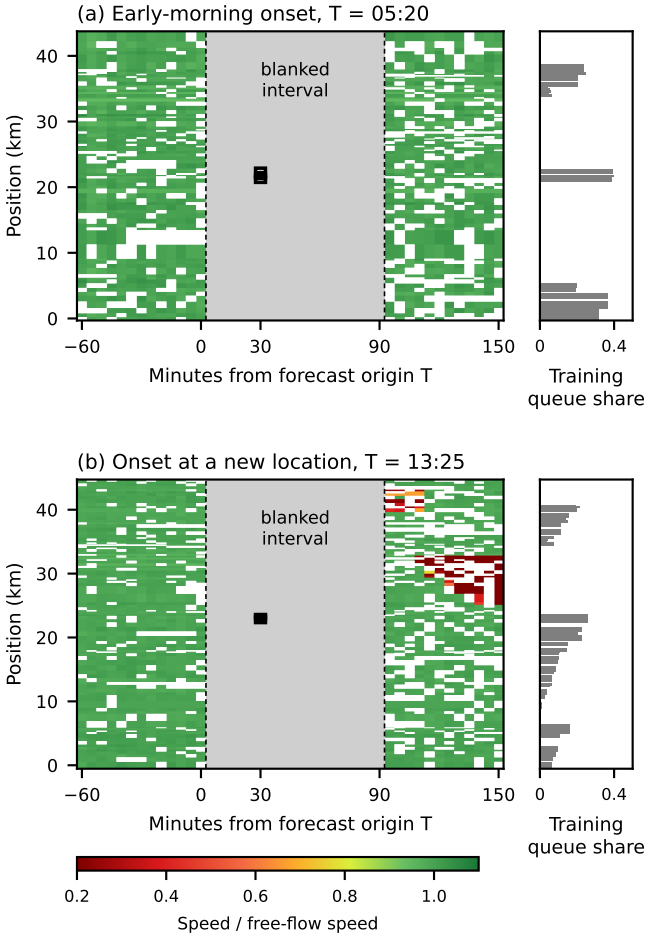


Fig. 5. Two April onset windows as time–space maps of speed relative to free-flow speed (white: withheld cells; grey band between dashed lines: blanked interval; black: links forecast as queued at $T+30$; right: each link's queue share in training). (a) An early-morning queue that has cleared an hour later. (b) Queues on links that rarely queued in training, away from the forecast bottleneck.

first. On the replicated March windows the location-agnostic ensemble ranked first, and the leaderboard agreed with the latter. A validation month generated like the evaluation months selected the model that transferred.

5.2 The pattern of missing data carries information

The information of corridor-wide blanked intervals sits in the interval's boundaries, the recurrent profile and indirect sensors such as on-ramp counts, so the way data go missing deserves its own model as much as the missing values do.

5.3 New queues are the remaining gap

From the public to the private month, onset IoU fell and ongoing IoU rose (Table II). The links our model forecast at $T+30$ were still queued an hour later in about two thirds of the onset windows in both months, so the loss most likely lies in onsets that the training months describe poorly: short early-morning queues, where a single link moves the IoU strongly, and queues on links that rarely queued in training (Fig. 5). April holds more of both.

5.4 Limitations

Two parts of the design learn where onsets occur from the training months, the link model's onset incidence and the bottleneck choice's candidates, and therefore favour familiar places. Incidents and demand steps that arrive after the forecast origin lie outside what the models observe. A forecast of the next demand step and incident information are the natural next inputs.

6. Conclusion

Our starting point was to find the parts of freeway traffic that stay the same from month to month. The physics of the bottleneck stays: capacity discharge, the timing of demand steps, and the groups of links that queue together. The place where queues form moves. Models that read the current state through these mechanisms, and that treat corridor-wide blanked intervals as a structure of the data, carried the solution with gradient-boosted trees on a laptop CPU. The same view points to the next step: describing demand and incidents directly, so that forecasts rely less on where queues formed before.

References

[1] J. Wu, "TrafficFlowBench: A four-task freeway traffic benchmark," version 1.0, Arizona State University, 2026. [Online]. Available: https://github.com/jacky850/trafficflowbench-public

[2] J. Wu and X. S. Zhou, "2026 IEEE Big Data-Traffic Flow Bench," Kaggle, 2026. [Online]. Available: https://www.kaggle.com/competitions/2026-ieee-big-data-traffic-flow-bench

[3] T. Seo, A. M. Bayen, T. Kusakabe, and Y. Asakura, "Traffic state estimation on highway: A comprehensive survey," *Annu. Rev. Control*, vol. 43, pp. 128–151, 2017.

[4] X. Chen, J. Yang, and L. Sun, "A nonconvex low-rank tensor completion model for spatiotemporal traffic data imputation," *Transp. Res. C*, vol. 117, Art. no. 102673, 2020.

[5] M. Yildirimoglu and N. Geroliminis, "Experienced travel time prediction for congested freeways," *Transp. Res. B*, vol. 53, pp. 45–63, 2013.

[6] B. Metzger, A. Loder, L. Kessler, and K. Bogenberger, "Spatio-temporal prediction of freeway congestion patterns using discrete choice methods," *EURO J. Transp. Logist.*, vol. 13, Art. no. 100144, 2024.

[7] W. Brilon, J. Geistefeldt, and M. Regler, "Reliability of freeway traffic flow: A stochastic concept of capacity," in *Transportation and Traffic Theory: Proc. 16th Int. Symp. (ISTTT)*, 2005, pp. 125–144.

[8] L. Breiman, "Stacked regressions," *Mach. Learn.*, vol. 24, no. 1, pp. 49–64, 1996.

[9] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Adv. Neural Inf. Process. Syst. 30*, 2017, pp. 3146–3154.

[10] M. Lumiste and A. Ilie, "Large scale traffic forecasting with gradient boosting, Traffic4cast 2022 challenge," arXiv:2211.00157, 2022.

[11] A. J. Huang and S. Agarwal, "On the limitations of physics-informed deep learning: Illustrations using first-order hyperbolic conservation law-based traffic flow models," *IEEE Open J. Intell. Transp. Syst.*, vol. 4, pp. 279–293, 2023.

[12] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: Unbiased boosting with categorical features," in *Adv. Neural Inf. Process. Syst. 31*, 2018, pp. 6638–6648.

[13] D. McFadden, "Conditional logit analysis of qualitative choice behavior," in *Frontiers in Econometrics*, P. Zarembka, Ed. New York, NY, USA: Academic Press, 1974, pp. 105–142.

[14] M. Treiber, A. Kesting, and R. E. Wilson, "Reconstructing the traffic state by fusion of heterogeneous data," *Comput.-Aided Civ. Infrastruct. Eng.*, vol. 26, no. 6, pp. 408–419, 2011.

[15] M. J. Lighthill and G. B. Whitham, "On kinematic waves II. A theory of traffic flow on long crowded roads," *Proc. R. Soc. Lond. A*, vol. 229, no. 1178, pp. 317–345, 1955.

[16] Y. Tashiro, J. Song, Y. Song, and S. Ermon, "CSDI: Conditional score-based diffusion models for probabilistic time series imputation," in *Adv. Neural Inf. Process. Syst. 34*, 2021, arXiv:2107.03502.

[17] A. Cini, I. Marisca, and C. Alippi, "Filling the g_ap_s: Multivariate time series imputation by graph neural networks," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022.

[18] X. Liu *et al.*, "LargeST: A benchmark dataset for large-scale traffic forecasting," in *Adv. Neural Inf. Process. Syst. 36, Datasets and Benchmarks Track*, 2023, arXiv:2306.08259.

[19] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why do tree-based models still outperform deep learning on typical tabular data?" in *Adv. Neural Inf. Process. Syst. 35, Datasets and Benchmarks Track*, 2022.

APPENDIX A.
DEVELOPMENT PATH

TABLE IV
EVERY SUBMISSION THAT RAISED THE BEST PUBLIC SCORE, AND THE FINAL FILE, WITH WHAT EACH CHANGED. Δ: CHANGE AGAINST THE PREVIOUS ROW. BOLD: THE SIX DECISIONS OF FIG. 4.

#	Date	Change	Public	Δ	Private	Δ
0	09-18	Baseline (Section 3.2)	0.78918		0.80736	
1	09-20	Task 2: serving features aligned with training; speed and occupancy dispersion features	0.79502	+0.00584	0.80366	−0.00370
2	09-20	**Task 1: learned stack of five reconstruction members**	0.80287	+0.00785	0.81200	+0.00834
3	09-21	Task 2: queue models refitted on all 273 training days	0.80397	+0.00110	0.81368	+0.00168
4	09-21	Ongoing: link threshold 0.475 → 0.675	0.80469	+0.00072	0.80983	−0.00385
5	09-21	Onset: link threshold 0.325 → 0.20	0.80881	+0.00412	0.81069	+0.00086
6	09-22	Onset: queue at $T+30$ alone; at least one link per window	0.81631	+0.00750	0.81631	+0.00562
7	09-23	**Onset: link model with onset-conditional incidence**; flow density projection, two corridors	0.82884	+0.01253	0.82353	+0.00722
8	09-23	**Ongoing: training windows sampled per condition**; density projection; shrunk onset thresholds	0.84377	+0.01493	0.83294	+0.00941
9	09-23	Ongoing: training windows cut by the replicated selection rule	0.84404	+0.00027	0.83941	+0.00647
10	09-24	**Task 1: gap model for blanked intervals**	0.86830	+0.02426	0.86270	+0.02329
11	09-24	**Ongoing: location-agnostic ensemble**	0.87481	+0.00651	0.87317	+0.01047
12	09-24	**Onset: breakdown risk per bottleneck**; group-level decoding for both conditions	0.88205	+0.00724	0.88594	+0.01277
13	09-24	Task 1: residual correction for scattered cells	0.88489	+0.00284	0.88896	+0.00302
14	09-25	Ongoing: blend with the selection-aware ensemble (0.75 / 0.25)	0.88596	+0.00107	0.88937	+0.00041
15	09-25	Task 1: second gap model; fallback for one empty ongoing window	0.88849	+0.00253	0.88954	+0.00017
16	09-25	Onset: conditional-logit bottleneck choice and queue-extent template	0.89974	+0.01125	0.89329	+0.00375
17	09-25	Task 1: per-month correction of gap speed	0.90033	+0.00059	0.89380	+0.00051
18	09-26	Task 1: two-state gap speed at weight 0.25	0.90038	+0.00005	0.89384	+0.00004
19	09-27	Task 1: queue probability shared over same-month groups, weight 0.5	0.90043	+0.00005	0.89390	+0.00006
20	09-27	Task 1: density-residual projection at strength 0.5	0.90058	+0.00015	0.89406	+0.00016
21	09-27	Task 1: CatBoost residual learner at half weight	0.90085	+0.00027	0.89436	+0.00030
22	09-29	Task 1: second residual layer on one corridor; on-ramp counts in the gap classifier	0.90112	+0.00027	0.89466	+0.00030
23	09-29	Ongoing: six selection-aware models refitted with January and February	0.90140	+0.00028	0.89454	−0.00012
24	09-30	Task 1: density refit below $0.85\,v_{\mathrm{f}}$	0.90144	+0.00004	0.89459	+0.00005
25	09-30	Ongoing: monotone ensemble	0.90187	+0.00043	0.89379	−0.00080
26	10-01	Ongoing: spacing correction of the selection-aware inputs	0.90216	+0.00029	0.89343	−0.00036
27	10-01	Ongoing: group threshold 0.45 (final submission)	0.90115	−0.00101	0.89493	+0.00150

APPENDIX B.
INFORMATION BOUNDARY OF TASK 2

Inputs. For a window with origin T, serving reads the window's released history from $T-60$ to $T-5$, the released mainline and ramp values at T, up to five earlier days of the same split, the previous official origin, and models and tables fitted on the training split; it is independent of the outputs of Tasks 1, 3 and 4.

Missing history cells. They stay missing: LightGBM routes them natively, the monotone ensemble substitutes training-split medians with a missing-value flag, and the queue-geometry features interpolate within the twelve history stamps.

Validation fill. The Task 1 fill of the replicated March histories (Section 3.1) uses both ends of each gap and the whole month, so it may carry information from after T; every compared variant sees the same fill, and the validation run serves relative comparisons. The ongoing threshold 0.45, chosen on a filled validation run, was confirmed on the unfilled one ($+0.0049$ IoU, 95% interval $+0.0012$ to $+0.0085$).

Choices. Task 2 models were fitted on the training split and its constants selected on the training months and released March observations, as the rules allow for April windows; April observations entered Task 2 at serving alone.

Truncation test. For a cut at the origin T of a chosen window, the package's `audit/` folder removes every row stamped after T from the masked mainline and ramp files, window histories and window index of both evaluation splits, serves Task 2 on that view, and compares every window with origin at or before T with the full run. Twelve cuts in both months and both window types compared 1,216 windows: all 1,342,530 queue decisions match, and in three cuts all 5,609,904 served probabilities match exactly. In three controls that also remove the rows stamped at T, every probability of the window at T moves (218, 18,600 and 15,066 values, by up to 0.14, 0.21 and 0.33), while the 246 earlier windows stay identical, so the test detects a single time stamp (`audit/RESULTS.md`).

No fitting at serving. A probe on every fitting call of LightGBM, scikit-learn, CatBoost, the SciPy optimisers and PyTorch recorded none in a full Task 2 run: every model loads from a frozen file, and the per-month inputs (recency, spacing to the previous window, replays of the selection rule) come from the released data of the month being served. Task 2 therefore runs on a new month in the released layout without refitting, in about 5 min on the laptop CPU.

APPENDIX C.
MODEL SETTINGS

TABLE V

SHIPPED MODELS: TRAINING DATA, SETTINGS, INPUTS, SEED AND RECORDED FITTING TIME. THE IN-RUN ROW IS FITTED ON EACH MONTH'S RELEASED OBSERVATIONS, EVERY OTHER MODEL ON THE TRAINING SPLIT, JUNE 2030 TO FEBRUARY 2031, ALL ON THE LAPTOP CPU OF SECTION 4. LR: LEARNING RATE; MIN: MINIMUM ROWS PER LEAF; FF/BF: FEATURE AND BAGGING FRACTIONS; λ_1, λ_2: L1 AND L2 PENALTIES; DEFAULT: SEED LEFT UNSET (0). UNLISTED LIGHTGBM PARAMETERS KEEP THEIR DEFAULTS (LIGHTGBM 4.7.0, CATBOOST 1.2.10). –: NOT RECORDED OR NOT APPLICABLE. TASK 4 FITS NO MODEL.

Model	Training data	Learner and settings	Inputs	Seed	Fit time
Traffic state (Tasks 1 and 3)					
Learned stack (speed, flow)	2030-06-01 to 12-20 (203 days), 10 corridors, 25% of target cells: 5.86 M rows	LightGBM L2; 127 leaves, lr 0.05, min 200, ff/bf 0.8; 900 rounds	71	20260920	1,191 s
Density model of the projection	2030-06-01 to 12-20, I-10 E and I-5 N (D12), 1,000 cells per day: 0.41 M rows	LightGBM L1 weighted by link length; 127 leaves, lr 0.05, min 200, ff/bf 0.8; 600 rounds	71	20260922	34 s
Gap model (speed, per-lane flow)	Blanked intervals at replicated windows, 8 corridors, 35% of cells; fit 2030-06 to 12 (3.60 M rows), early stopping 2031-01 (0.55 M), test 2031-02	LightGBM L2; 127 leaves, lr 0.05, min 200, ff/bf 0.8, λ_2 1; 3,000 rounds	73	1	2,119 s
First gap model	Blanked intervals at replicated windows, 8 corridors	LightGBM L2; λ_2 1; 2,984 / 3,000 trees	51	1	about 9 min
Residual LightGBM	2031-01-04 to 02-28, 8 corridors; free-flow cells sampled at 30% with weight 1/0.3: 1.98 M rows	LightGBM L2; 63 leaves, lr 0.05, min 300, ff/bf 0.8, λ_2 10; 400 rounds	130	default	314 s
CatBoost residual	Rows of the residual LightGBM	CatBoost RMSE; depth 6, lr 0.05, L2 10, 128 borders, MVS subsample 0.8; 4,800 iterations	130	0	3,102 s
Density heads (all cells; below $0.85\,v_{\mathrm{f}}$)	2031-01-04 to 01-31, 2.93 M cells; 2031-02, 3.02 M cells; parents out of fit	LightGBM L2; 31 leaves, lr 0.05, min 400, ff/bf 0.8, λ_2 10; 200 rounds	143	default; 0	39 s; 62 s
Second residual layer, I-405 (D12)	January fit, February test	LightGBM L2; 31 leaves, lr 0.05, min 300, ff/bf 0.8, λ_2 10; first 100 of 200 trees	106	0	40 s
Ramp residual of interval speed	2031-01 intervals at replicated windows, 50% of cells: 0.79 M rows	LightGBM L2; 31 leaves, lr 0.05, min 500, ff/bf 0.8, λ_2 10; 500 rounds	57	5	–
Month correction, two-state heads, ramp queue probability	Pseudo intervals cut into each month's released observations, fitted within the run	LightGBM; 31 leaves, lr 0.05, min 400 (ramp head 500), ff/bf 0.8, λ_2 10; 300 rounds (ramp head 400)	–	7 (ramp head 9)	in run
New queues (Task 2 onset)					
Link model	2,101 replicated onset windows; fit to 2030-12-31, early stopping 2031-01: 307,077 link rows, window-balanced weights	LightGBM binary; 127 leaves, lr 0.1, min 10, bagging 0.8 every 5 rounds, λ_1 8.544, λ_2 0.0958, 127 bins; 129 trees	75	20260922	6.8 s
Breakdown risk (12 models)	273 training days, 4 corridors, three of them served; every origin 05:00–20:45 with the bottleneck free from $T-60$ to $T+25$	Standardised logistic regression, $C = 1$, L-BFGS; one model per bottleneck	12	deterministic	–
Bottleneck choice (24 logits)	2,108 replicated onset windows, or all training origins with a queued bottleneck; per corridor	Conditional logit; 4 specifications $\times$ 2 training sets $\times$ L2 penalty $\{1, 5, 20\}$; L-BFGS-B, at most 2,000 iterations	per spec.	deterministic	–
Existing queues (Task 2 ongoing)					
Location-agnostic ensemble (6)	Replicated ongoing windows: fit to 2030-12-31 (2,352), early stopping 2031-01 (354)	LightGBM binary, corridor-balanced weights; 127 leaves, lr 0.05, min 100, ff 0.9, bf 0.8, λ_2 10, 127 bins; 490–697 trees	73, 93	20260922, 20260929, 20260936	–
Selection-aware ensemble (6)	3,013 replicated windows to 2031-02-28: 2.58 M rows; selection inputs blanked in 765 windows	Same recipe; fixed 554–693 rounds	78, 98	20260922 + {0,1,2}	121–183 s each
Monotone ensemble (6)	The same 2.58 M rows	Same recipe with 29 monotone constraints (intermediate method); 554, 577, 681 rounds	156, 196	20260922–24	113–215 s each

APPENDIX D.
FIXED CONSTANTS

TABLE VI

FIXED CONSTANTS OF THE FINAL SUBMISSION, THE DATA ON WHICH EACH WAS SET, AND THE RECORDED COMPARISON. QUEUE ROWS: CHANGE IN ONSET OR ONGOING IOU ON REPLICATED WINDOWS (SECTION 3.1); TASK 1 ROWS: CHANGE IN THE LOCAL STATE-AND-PHYSICS SCORE OR IN THE INTERVAL-SPEED MEAN-SQUARED ERROR (MSE). TASK 1 RUNS OFFLINE, AND ITS GATES ALSO READ THE RELEASED MARCH AND APRIL OBSERVATIONS.

Constant	Value	Set on	Recorded comparison
Co-switching group cut	0.5	Training months (ongoing decoding: to 2031-01-31)	Any cut in 0.2–0.8 moves at most 14% of pair boundaries
Onset link thresholds	0.30; per corridor 25% local, 75% global (log-odds)	217 replicated February onset windows	Shrinkage 25%: $+0.0007$ in 3 of 3 forward folds; 50%: -0.0012
Onset group threshold	0.7	Replicated March windows	0.5 / 0.6 / 0.7 against link thresholds: $+0.029$ / $+0.040$ / $+0.051$
Bottleneck-choice ensemble	Mean of 24 conditional logits	Replicated March windows	Single configurations -0.0009 to $+0.0327$ (22 of 24 positive, median $+0.0129$); mean $+0.0262$
Ongoing ensemble weights	0.75 / 0.25	Replicated March windows; rechecked	Location-agnostic weight 0.15 / 0.35: -0.0003 / -0.0005
Monotone share per slot	0.5	Replicated March windows	0.25: -0.0010 (filled validation run); 0.75 / 1.0: -0.0026 / -0.0127
Ongoing group threshold	0.45	Odd replicated March days; confirmed on even days	0.40 / 0.45 / 0.55 / 0.60 against 0.50: -0.0038 / $+0.0011$ / -0.0022 / -0.0079; even days $+0.0079$ (95% interval $+0.0040$ to $+0.0167$)
Density-residual projection	Strength 0.5, L2 head	February 2031, six candidates	L1 or L2 head at 0.1 / 0.25 / 0.5: $+0.000015$ to $+0.000141$
CatBoost residual weight	0.5	Late January; confirmed in February and March	Weight 1: $+0.00022$; weight 0.5: $+0.00038$
Density refit gate	Speed below $0.85\,v_{\mathrm{f}}$	February; March and April gates	Gated against every cell: $+0.00001$ (February)
Two-state weight and groups	0.5; same-month clusters (queue Jaccard $\geq$0.95)	March pseudo intervals; April gate	Interval-speed MSE: no groups -1.12%, training-month groups -0.91%, same-month groups at 0.25 -1.29%, at 0.5 -3.55% (April -3.78%)

APPENDIX E.
ALTERNATIVES COMPARED

TABLE VII

ALTERNATIVES TO THE RETAINED CHOICES, FROM PAIRED LEADERBOARD SUBMISSIONS AND LOCAL COMPARISONS ON THE PROTOCOLS OF SECTION 3.1. LEADERBOARD ROWS: PUBLIC / PRIVATE CHANGE OF S_{total} AGAINST THE PARENT SUBMISSION, WHICH DIFFERS IN THAT PART ALONE.

Part	Alternative	Measure	Result
Task 1	Temporal smoothing of the reconstructed field	Leaderboard	-0.00418 / -0.00400
Onset	Link threshold 0.65 (retained then: 0.325)	Leaderboard	-0.00899 / -0.01198
Onset	Link threshold 0.15 (retained then: 0.20)	Leaderboard	-0.00120 / $+0.00041$
Onset	Top link alone where the best probability is below 0.5	Leaderboard	-0.00312 / 0.00000
Task 1	Free- and queued-state gap heads reading on-ramp counts	Leaderboard	-0.00001 / $+0.00007$
Task 1	Conservation term in the loss (joint speed–flow network)	February, state-and-physics score	-0.000085
Task 1	Interval speed from nearest-neighbour segments; duration bridges (HMM, HSMM)	Pseudo intervals, interval-speed MSE	$+15.7\%$ / $+22.9\%$; $+1.6\%$, $+2.6\%$ (retained: -3.55%)
Task 2	Sequence network along the links (temporal convolution)	Local queue score, same cells and folds	0.685 against 0.846 for LightGBM
Task 2	Focal, Lovász-hinge and class-balanced losses	Out-of-fold queue score	-0.060, -0.027, -0.046
Onset	Historical frequency as fixed starting log-odds	January / February onset score	0.829 / 0.876 against 0.865 / 0.897
Onset	Bottleneck with the largest q/c	Replicated March, onset IoU	-0.090 to -0.100 against the risk choice
Onset	One pre-registered logit; one logit pooled over corridors	Replicated March, onset IoU	$+0.0056$; -0.039 (mean of 24: $+0.0262$)
Ongoing	CatBoost, XGBoost, ExtraTrees; TabM (32 members)	January selection; February / March	All at or below the LightGBM parent; TabM -0.0127 / $+0.0064$
Ongoing	Expected-IoU decoders; 12 seeds per member	Replicated March, ongoing IoU	-0.0039 to -0.0047, copula 5 of 5 negative; seeds -0.0008 / -0.0019
Task 4	Multiplicative spline prior on trips	Held-out counts, WAPE (training split)	0.26 $\rightarrow$ 0.16, with total path flow halved